



Cu-doped TiO₂ nanoparticles for photocatalytic disinfection of bacteria under visible light

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ABSTRACT

Two percent Cu-doped TiO₂ nanoparticles were prepared by a modified ammonia-evaporation-induced synthetic method, calcined at 450 °C, and characterized by powder X-ray diffraction, energy dispersive X-ray analysis, ESR spectroscopy, scanning electron microscopy, UV–visible diffuse reflectance spectrum, photoluminescence spectroscopy, and electrochemical impedance spectroscopy. Doping shifts the optical absorption edge to the visible region but increases the charge-transfer resistance and decreases the capacitance. Under visible light, the composite nanoparticles very efficiently catalyze the disinfection of *Escherichia coli*. The prepared oxide is selective in photocatalysis; under UV light, its photocatalytic activity to degrade sunset yellow, rhodamine B, and methylene blue dyes is less than that of the undoped one.

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1. Introduction

Illumination of semiconductor nanocrystals with light of energy not less than the band gap creates electron–hole pairs, electrons in the conduction band and holes in the valence band [1,2]. A fraction of these pairs diffuse to the crystal surface and react with the adsorbed substrates, resulting in photocatalysis, and the rest recombine, lowering the photocatalytic efficiency. While the hole oxidizes the organics the adsorbed oxygen molecule takes up the electron and transforms into highly active superoxide radical (O₂^{•−}). In the presence of moisture, O₂^{•−} produces reactive species such as HO[•], OH₂[•], and H₂O₂, which also oxidize the organics. Water is adsorbed on the semiconductor surface, molecularly and dissociatively. Hole trapping by either the surface hydroxyl groups or the adsorbed water molecules produces short-lived HO[•] radicals, which are the primary oxidizing agents. Semiconductor photocatalysis gains interest owing to its application in environmental remediation and TiO₂ is a promising material for photocatalytic application owing to its exceptional optical and electronic properties, chemical stability, nontoxicity, and low cost [1–3]. Although rutile TiO₂ absorbs in the visible region (~410 nm) it is less photocatalytically active than anatase, which is activated by UV-A light (~385 nm) [4,5]. Anatase combined with rutile shows better photocatalytic activity due to enhanced separation of photogenerated electron–hole pairs [5,6]. TiO₂ P25 Degussa (ca. 80% anatase, 20% rutile), a benchmark photocatalyst, shows excellent activity due

to a synergistic effect between anatase and rutile phases. The charge produced on rutile by visible light is stabilized through rapid electron transfer from rutile to anatase and the photoactivity is extended to the visible region [5]. Potable water is a basic essential requirement and microbial contamination and growth in surface water are potential health hazards. A simple disinfection method adoptable even in inaccessible regions of developing countries is the current need. World Health Organization's SODIS (Solar water disinfection) is a solution but it requires bright sunlight for 6 h. Semiconductor photocatalysis is a possible alternative for point-of-use water disinfection. TiO₂ photocatalytically disinfects bacteria but under UV light [7–9]. Cu–TiO₂ [10], Pd–TiO₂, and Pd–SnO₂ [11] thin films also require UV-A light for photodisinfection; *Escherichia coli* (*E. coli*) are the bacteria employed as biological indicators to evaluate the disinfection efficiency of the photocatalytic sterilization. Fe³⁺-doped TiO₂ thin film, prepared by a sol–gel technique, also needs UV light to kill *E. coli* [12]. Photocatalytic disinfection of *E. coli* by TiO₂–Fe₂O₃ [13], TiO₂–NiFe₂O₄ [14], Nd³⁺-doped TiO₂–NiFe₂O₄ [15], and W⁴⁺-doped TiO₂–NiFe₂O₄ [16] composites has been reported but under germicidal UV-C light. Disinfection of *E. coli* with Ag–TiO₂ has been studied under UV light [17–20]; however, Ag-doped TiO₂ is a bactericide and inactivates *E. coli* in dark itself [21,22]. Doping TiO₂ with nitrogen and/or sulfur shifts its optical edge to the visible region and *E. coli* disinfection with visible light has been reported [23–26]. While the N- and S-codoped TiO₂ requires about 1.25 h to disinfect *E. coli* of 10⁴ CFU mL^{−1} under blue light (400–500 nm) [23–25], the N-doped TiO₂ needs 2 h for inactivation of *E. coli* of population 10⁹ CFU mL^{−1} [26]. Here we report for the first time very efficient

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photocatalytic disinfection of bacteria by Cu-doped TiO₂ under visible light; TiO₂ P25 is the precursor of the composite oxide prepared.

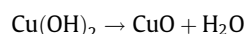
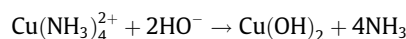
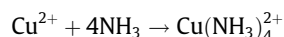
2. Materials and methods

2.1. Materials

TiO₂ P25 was a gift from Degussa. MacConkey agar (Himedia) and nutrient broth (SRL) were used as received. Methylene blue (Sd fine), rhodamine B (Sd fine), and sunset yellow (Sigma–Aldrich) were used as supplied. Other chemicals used were also of analytical or reagent grade. Deionized and doubly distilled water was employed throughout the experiments.

2.2. Photocatalyst preparation

Two percent Cu-doped TiO₂ nanoparticles were prepared by a modified ammonia-evaporation-induced synthetic method [27]. Cu(NH₃)₄²⁺ complex cation was prepared in situ by addition of ammonia to TiO₂ P25, suspended in a required volume of millimolar Cu²⁺ solution under vigorous stirring, to reach a pH of 11.4. Development of blue coating over the TiO₂ nanocrystals indicated the formation of a copper(II) amine complex. Continuous stirring for 24 h ensured uniform coating. Evaporation to dryness resulted in the conversion of the complex to hydroxide. The composite nanoparticles were obtained by calcination at 450 °C for 4 h in a muffle furnace fitted with a PID temperature controller and the heating rate was set at 10 °C min^{−1}.



2.3. Characterization techniques

The powder X-ray diffractogram (XRD) of the catalyst was recorded with a Bruker D8 system using Cu K α radiation of wavelength 1.5406 Å in a 2 θ range of 5–75° at a scan speed of 0.050° s^{−1}. A JEOL JSM-5610 scanning electron microscope (SEM) equipped with BE detector was used to determine the morphology of the sample. The sample was placed on an adhesive carbon slice supported on copper stubs and coated with 10-nm-thick gold using a JEOL JFC-1600 auto fine coater prior to measurement. The energy dispersive X-ray (EDX) spectrum was recorded with a JEOL JSM-5610 SEM equipped with EDX. The ESR spectrum was recorded at 9.5 GHz on X band with a JEOL-JES-TE 100 ESR spectrometer. A Cary 500 spectrophotometer was employed to record the UV–visible diffuse reflectance spectrum of the oxide. The photoluminescence (PL) spectra were obtained using a Perkin Elmer LS 55 fluorescence spectrometer at room temperature. The nanocrystals were dispersed in carbon tetrachloride and the wavelength of excitation was set at 250 nm. The electrochemical impedance spectra (EIS) were obtained using a HP 4284A Precision LCR meter over the frequency range of 1 MHz to 20 Hz at room temperature in air; the disk area was 0.5024 cm² and the thicknesses of Cu-doped TiO₂ and TiO₂ pellets were 2.47 and 2.94 mm, respectively.

2.4. Photoreactors

An immersion-type photoreactor with a 150 W tungsten halogen lamp fitted into a double-walled borosilicate immersion well of 40 mm outer diameter with inlet and outlet for water circulation

was employed for visible light photocatalysis. The reaction vessel was a 100 mL borosilicate immersion well of 50 mm outer diameter. Fig. 1 is the emission spectrum of the light source used. For UV photocatalysis a multilamp photoreactor fitted with 8 W mercury lamps of wavelength 365 nm (Sankyo Denki, Japan) and a highly polished anodized aluminum reflector was employed. The reaction vessel was borosilicate glass tube of 15 mm inner diameter and was placed at the center of the reactor. The cooling fans at the bottom of the reactor dissipate the heat generated. The reactor was illuminated by four lamps mutually set at right angles. The photon flux of the UV light (*I*) was determined by ferrioxalate actinometry.

2.5. Bacterial culture

A nutrient broth culture medium of pH 7.4 was prepared by dissolving 13.0 g nutrient broth (5.0 g peptone, 5.0 g NaCl, 2.0 g yeast extract, 1.0 g beef extract) in 1 L distilled water followed by sterilization in an autoclave at 121 °C. MacConkey agar plates were prepared separately by dissolving 55 g MacConkey agar (20 g peptic digest of animal tissue, 10 g lactose, 5 g sodium taurocholate, 0.04 g neutral red, 20 g agar) in 1 L boiling distilled water followed by sterilization in an autoclave at 121 °C and poured into petri dishes. *E. coli* bacteria were inoculated in 10 mL of a nutrient broth and incubated for 24 h at 37 °C. Ten milliliters of the grown culture was diluted with nutrient broth to 100 mL and used for the photocatalytic experiments. For the counting of *E. coli* colonies in CFU mL^{−1}, the bacterial solution was successively diluted to 10⁸–10¹² times with the nutrient broth. Ten microliters of the diluted *E. coli* was streaked on a MacConkey agar plate using a loop and incubated at 37 °C for 24 h. The CFU was counted by a viable count method.

2.6. Photocatalytic studies

Solutions of *E. coli* or the dyes of desired concentrations were prepared afresh and used. The volume of *E. coli* solution illuminated with tungsten lamp was 50 mL, whereas that of the dyes with mercury lamps was 25 mL. After the addition of the catalyst to the *E. coli* or dye solution, air was bubbled through the solution that kept the catalyst particles under suspension and at constant motion. The dissolved oxygen was measured using an Elico dissolved oxygen analyzer PE 135. After the illumination, the catalyst was separated. After proper dilution, the leftover *E. coli* was determined by a viable count method. The dyes, after dilution, were analyzed spectrophotometrically. The absorbance values at 662, 482, and 553 nm were used to estimate methylene blue, sunset

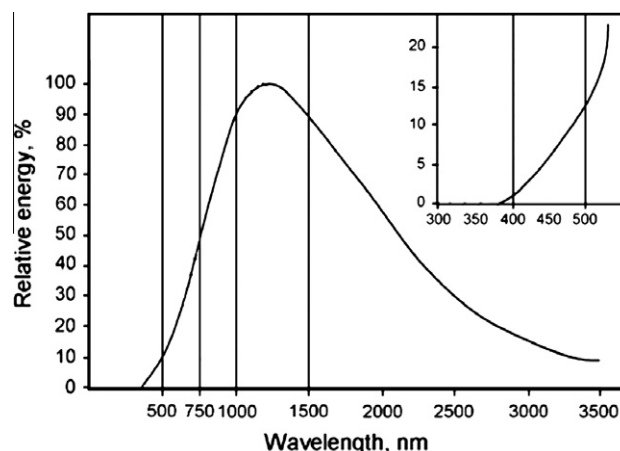


Fig. 1. Emission spectrum of W-lamp.

yellow, and rhodamine B, respectively. A calibration curve for each dye was constructed by measuring the absorbance at different parts per million.

3. Results and discussion

3.1. Photocatalyst characterization

Fig. 2 displays the X-ray diffraction pattern of the Cu-doped oxide. It confirms the presence of both anatase and rutile phases of TiO_2 in the sample. The standard JCPDS patterns of anatase (00-021-1272 (*), tetragonal, body centered, $a = b = 3.7852 \text{ \AA}$, $c = 9.5139 \text{ \AA}$, $\alpha = \beta = \gamma = 90.0^\circ$) and rutile (01-075-1750 (D), tetragonal, primitive, $a = b = 4.5937 \text{ \AA}$, $c = 2.9587 \text{ \AA}$, $\alpha = \beta = \gamma = 90.0^\circ$) match with the recorded XRD. However, the diffraction pattern of CuO is not seen in the XRD of the doped material. This is likely due to a low composition of Cu. The phase percentages of anatase and rutile have been deduced from the integrated intensity of the peaks at 2θ value of 25.3° for anatase and 27.4° for rutile. The percentage of anatase is given by $A (\%) = 100 / \{1 + 1.265(I_R/I_A)\}$, where I_A is the intensity of the anatase (1 0 1) peak at $2\theta = 25.3^\circ$ and I_R is that of the rutile (1 1 0) peak at $2\theta = 27.4^\circ$. The phase composition thus obtained is 79% anatase and 21% rutile, which is very close to that of the precursor TiO_2 P25 (81% anatase, 19% rutile). The phase composition of the precursor TiO_2 P25 has also been determined from its recorded powder XRD pattern (not shown). The average crystal sizes of the nanoparticles have been deduced from the half-width of the full maximum (HWHM) of the most intense peaks of the doped oxide and its precursor TiO_2 P25 using the Scherrer equation, $D = 0.9\lambda / \beta \cos \theta$, where D is the mean crystallite size, λ is the X-ray wavelength, θ is the Bragg angle, and β is the corrected line broadening of the sample. The specific surface areas of the nanocrystals have been obtained using the relationship, $S = 6/\rho D$, where S is the specific surface area, D is the average particle size, and ρ is the material density. The results are presented in Table 1. The mean crystallite size of Cu-doped TiO_2 is slightly larger than its precursor and this is because of the nanodeposition of the dopant on TiO_2 crystals. The slight increase in the crystal size on doping with Cu leads to a small decrease in the surface area. The energy dispersive X-ray spectroscopic analysis of the doped oxide

Table 1

Crystal size (D), surface area (S), ohmic (R_Ω) and charge-transfer (R_{CT}) resistances, specific conductance (σ), and capacitance (C).

Oxide	D , nm	S , $\text{m}^2 \text{g}^{-1}$	R_Ω , $\text{k}\Omega$	R_{CT} , $\text{k}\Omega$	σ , $\mu\text{S m}^{-1}$	C , pF
Cu-doped TiO_2	23.6	64.0	43.9	2980	16.5	6.7
TiO_2	22.7	66.6	4.0	36.8	1500	43.3

confirms the presence of Cu in the prepared oxide and the counted percentage of Cu is 2, which agrees with the stoichiometry. Fig. 3 presents the EDX spectrum of the doped oxide. The ESR spectrum of the doped oxide, recorded at room temperature and presented in Fig. 4, does not exhibit hyperfine splitting but displays a broad resonance yielding the g value as 2.217 which corresponds to CuO [28]. The SEM image of the composite oxide, displayed in Fig. 5, shows the sample as polycrystalline. The particles are highly agglomerated.

The diffuse reflectance spectrum of the Cu-doped TiO_2 is shown in Fig. 6. The reflectance data, reported as $F(R)$ values, have been obtained by application of the Kubelka–Munk algorithm. The band gap of the doped oxide has been deduced from the Tauc plot. Fig. 7 is the plot of $[F(R)h\nu]^{1/2}$ versus photon energy. The extrapolation of $[F(R)h\nu]^{1/2}$ to the abscissa at zero $F(R)$ provides the band gap energy as 2.83 eV, which corresponds to an absorption edge of 438 nm. The measured reflectance data of the doped oxide fit more satisfactorily the $[F(R)h\nu]^{1/2}$ versus photon energy plot than the $[F(R)h\nu]^2$ versus photon energy plot, suggesting a strong indirect band gap transition. Fig. 8 presents the photoluminescence spectrum of the prepared oxide. The PL spectrum of the undoped oxide is also displayed for comparison. Both spectra show near band gap emission (NBE) and blue or deep level emission (DLE). The NBEs of Cu– TiO_2 and TiO_2 at 418 nm correspond to the direct band gaps of the oxides. The DLE of both oxides at 485 nm is likely due to crystal defects, most probably arising out of the oxygen vacancies in the lattice [29]. The PL spectra reveal suppression of the NBE on doping TiO_2 with Cu. Also, the relative PL intensity of the NBE to the DLE ($I_{\text{NBE}}/I_{\text{DLE}}$) reduces from 1.3 to 1.2 on doping TiO_2 with Cu. This may be because of suppression of recombination of the photogenerated electron–hole pairs on doping TiO_2 with Cu. Creation of heterojunction slows down the electron–hole recombination and hence

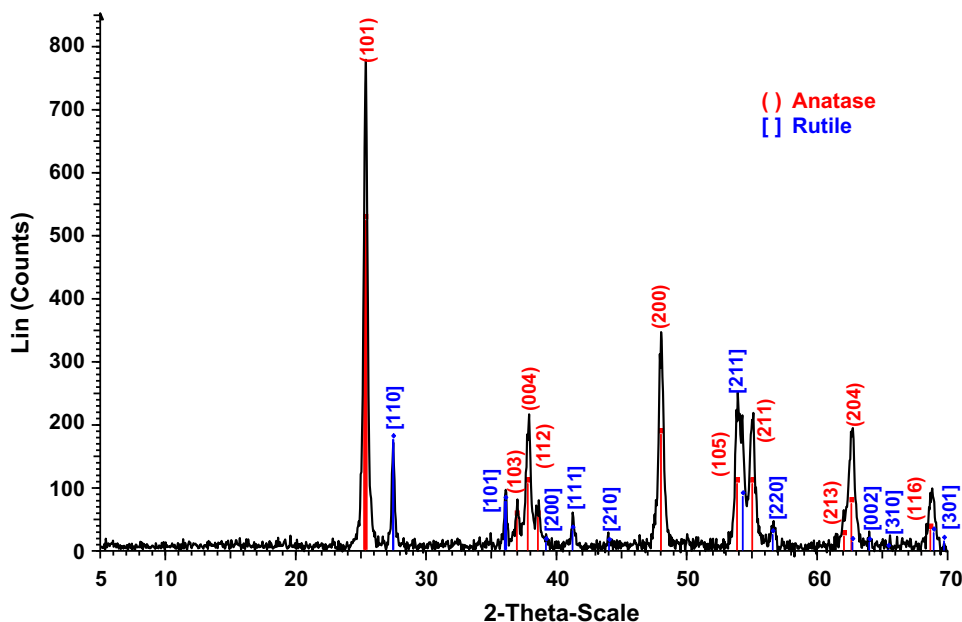


Fig. 2. X-ray diffractogram of the composite oxide.

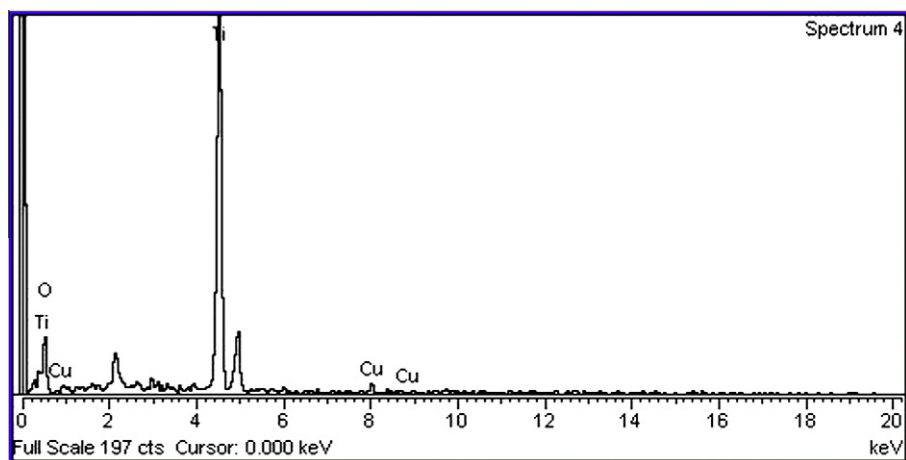


Fig. 3. EDX spectrum of the composite oxide.

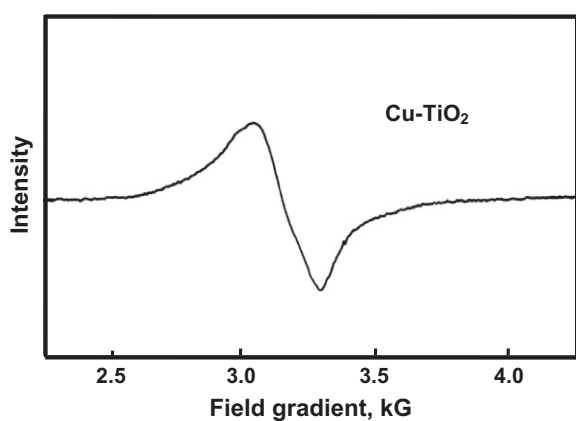


Fig. 4. ESR spectrum of the composite oxide.

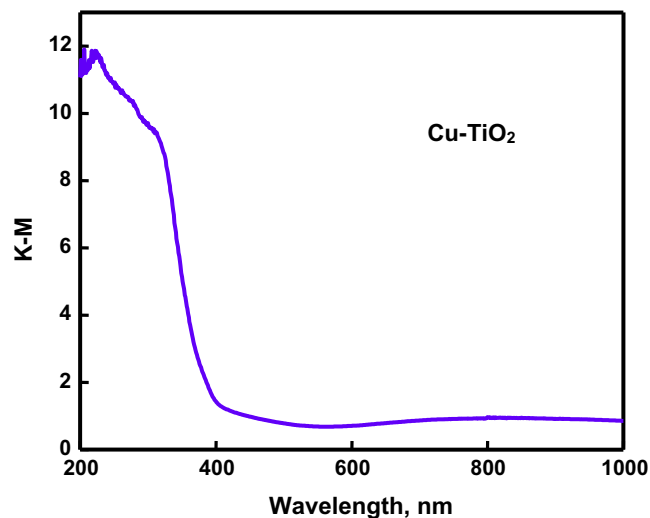


Fig. 6. DRS of the composite oxide.

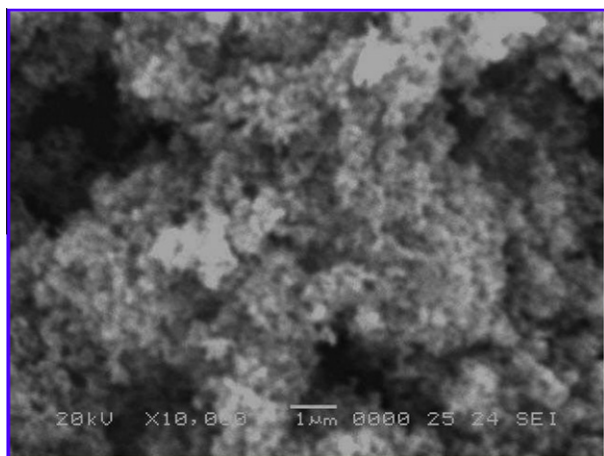
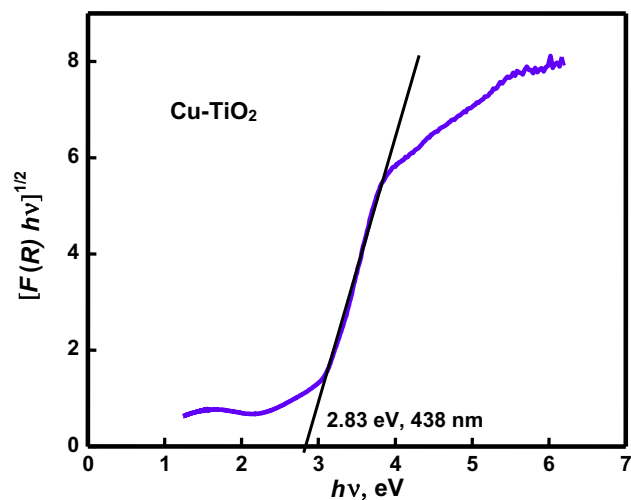
Fig. 5. SEM image of Cu-doped TiO₂.

Fig. 7. Tauc plot of the doped oxide.

reduces the emission. It is pertinent to state that the conduction and valence bands refer to the reduced and oxidized states in the semiconductor. The conduction band (CB) electron corresponds to the reduced form of Ti⁴⁺ or Cu²⁺ (i.e., Ti³⁺ or Cu⁺) and the valence band (VB) hole refers to the oxidized form of O²⁻ (i.e., O⁻). The band gap excitation is the photoexcitation of an electron from O²⁻ to Ti⁴⁺ or Cu²⁺ and the reversion of the electron from Ti³⁺ or Cu⁺ to O⁻ is the electron-hole recombination. The low NBE of Cu-doped TiO₂

reveals better separation of photogenerated electron-hole pairs in the doped oxide. The CB of CuO is less cathodic ($E_{CB} = -4.96$ eV) than that of TiO₂ ($E_{CB} = -4.21$ eV) [30]. Hence the photogenerated

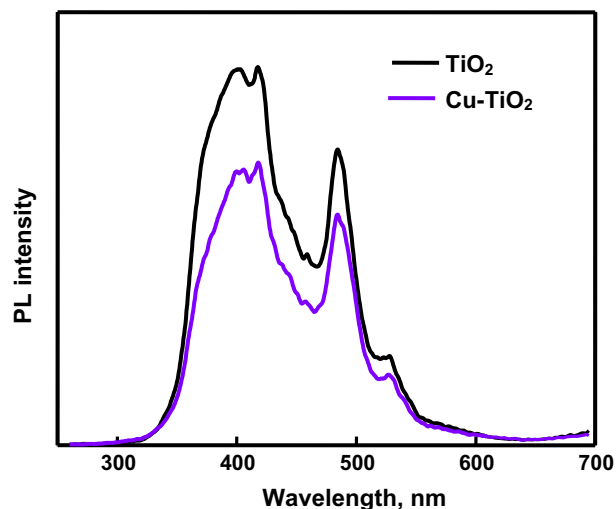


Fig. 8. The PL spectra.

electron in the CB of TiO_2 may slip to the CB of CuO . That is, Cu^{2+} may accept an electron to form Ti^{3+} . Similarly, the valence band of TiO_2 is more anodic ($E_{\text{VB}} = -7.41 \text{ eV}$) than that of CuO ($E_{\text{VB}} = -6.66 \text{ eV}$). This may also lead to hole transfer from the VB of TiO_2 to the VB of CuO . Another salient feature of the PL spectrum of the doped oxide is that it does not show any emission near 438 nm, which corresponds to the indirect band gap, deduced from DRS studies. In indirect band gap transition, electron from the CB recombines with hole in VB indirectly through traps without emission of photon.

Electrochemical impedance spectroscopy is a relatively new and powerful tool for probing the electrical properties of semiconductors [31]. It could be used to investigate the dynamics of the mobile and bound charges in the interfacial or bulk region of the semiconductors. In polycrystalline materials, the overall sample resistance may be a combination of the intragranular or bulk crystal resistance and intergranular or grain boundary resistance. Generally, the impedance data are analyzed in terms of an equivalent circuit model. An electrode interface undergoing an electrochemical reaction is typically analogous to an electric circuit consisting of a specific combination of capacitors and resistors. By fitting the EIS data to a model or an equivalent circuit the electrical properties of the semiconductors could be inferred. Measurement of impedance

of Cu-doped TiO_2 and undoped TiO_2 shows a decrease of impedance with an increase of frequency, indicating the capacitance of the semiconductors studied. Fig. 9 presents the Nyquist plot, which is a popular format for evaluating the impedance data. The semicircle in the Nyquist plot is the expected response of the simple circuit [31]. The ohmic or uncompensated resistance (R_{Ω}) corresponds to the grain boundary or intergranular resistance and the polarization or electron-transfer (charge-transfer) resistance (R_p or R_{CT}) refers to the intragranular or bulk crystal resistance. The R_{CT} is related to the Warburg resistance, which is the resistance to mass transfer. It is controlled by the specific conductance, σ . The constant phase element (CPE) is associated with a non-uniform distribution of current due to material heterogeneity and is equivalent to a double layer capacitance (C). Although Cu-doped TiO_2 is a composite, its Nyquist plot is a single semicircle, indicating the nonheterogeneity of the doped oxide. Table 1 presents the determined specific conductance, capacitance, and ohmic and charge-transfer resistances of Cu-doped TiO_2 and undoped TiO_2 . The specific conductance has been deduced from the measured charge-transfer resistance and the capacitance has been obtained using the equations $\omega_{\text{max}} = 1/CR_{\text{CT}}$ and $\omega_{\text{max}} = 2\pi f$, where f is the frequency corresponding to the maximum in the semicircle of the Nyquist plot. Cu doping decreases the specific conductance and capacitance.

3.2. Disinfection of *E. coli* under visible light

Fig. 10 displays the temporal profiles of photocatalytic disinfection of *E. coli* by Cu-doped TiO_2 , under tungsten lamp illumination or under natural AM1 sunlight. The corresponding profiles of undoped TiO_2 are also shown for comparison. The temporal profiles of *E. coli* inactivation by photolysis (without catalyst) are also provided in Fig. 10. The inactivation of the bacteria by Cu-doped TiO_2 in the dark is small compared to the photocatalytic disinfection (Fig. 10). The disinfection under natural sunlight was made with identical solar irradiance by performing the experiments simultaneously, side by side, in summer; the solar irradiance was monitored with a Daystar solar meter (USA). *E. coli* grown in a nutrient broth culture medium were used for the evaluation of the bactericidal activity. The cell population was determined by a viable count method on MacConkey agar plates, after proper dilution of the culture. Fig. 10 clearly demonstrates the effective photocatalytic disinfection of *E. coli* by Cu- TiO_2 under visible light. The observed initial rapid fall followed by a slow diminishing of the

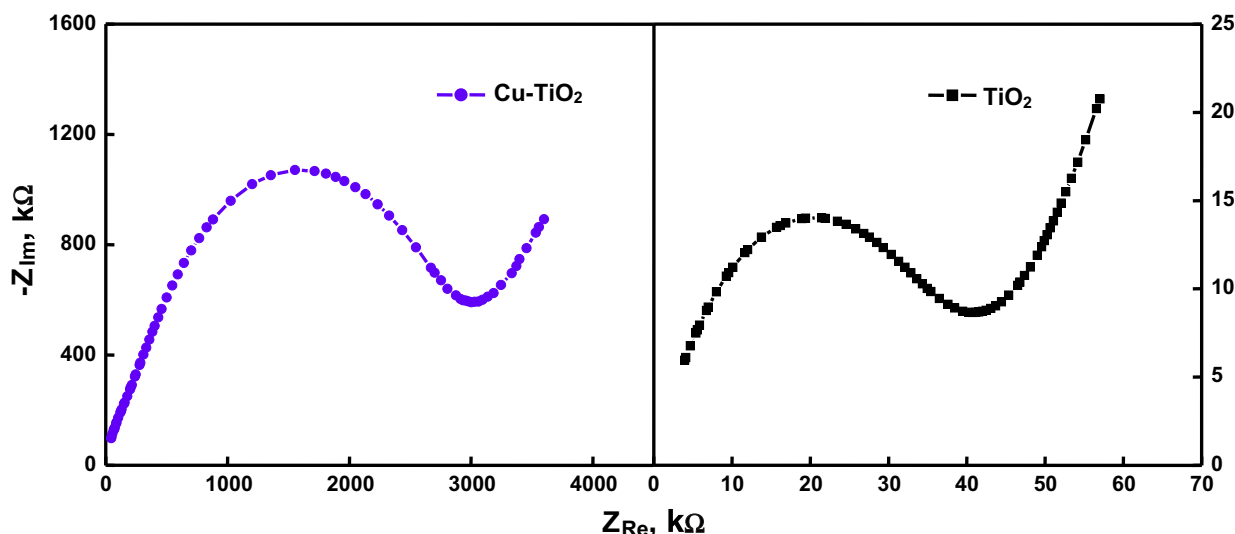


Fig. 9. The Nyquist plots.

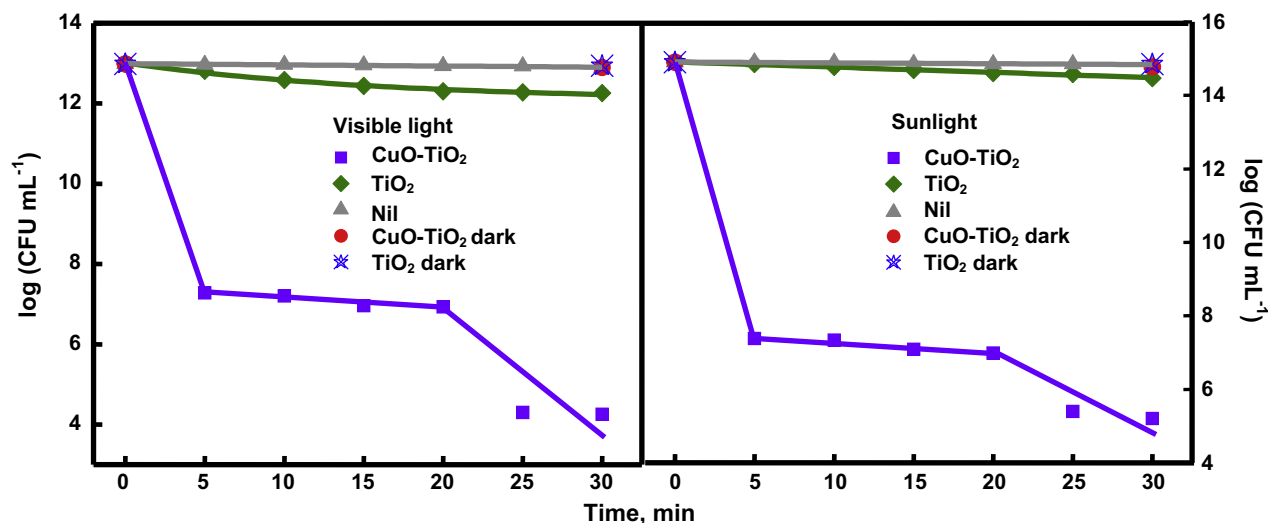


Fig. 10. Temporal profile of *E. coli* disinfection. Catalyst loading = 0.020 g, pH 7.1, *E. coli* solution = 50 mL, airflow rate = 7.8 mL s⁻¹, [O₂]_{dissolved} = 2.6 mg L⁻¹. Tungsten lamp illumination: light intensity = 1650 W m⁻². Natural sunlight: solar irradiance = ~1000 W m⁻²; illumination cross section = 11.43 cm².

E. coli population is similar to the results observed with TiO₂-Ni-Fe₂O₄ [14] and Ag-TiO₂ [17] nanocomposites, illuminated at 270 and 280 nm, respectively.

3.3. Selective photocatalysis

Reports on tuning TiO₂ to absorb visible light and demonstrating the photocatalytic activity of the tuned oxide to degrade a dye and thereby universalizing its photocatalytic efficiency are numerous. But our results clearly demonstrate against generalization of the photocatalytic efficiency, irrespective of the nature of the pollutant. The selectivity in photocatalysis of the prepared Cu-TiO₂ has been evaluated by employing three dyes of different structures and types. The dyes used are xanthen dye rhodamine B (C.I. 45,170), azo dye sunset yellow (C.I. 15,985), and heterocyclic dye methylene blue (C.I. 52,015). As these dyes absorb visible light the photocatalytic efficiency of the composite oxide has been evaluated using UV light of wavelength 365 nm. Fig. 11 presents the photodegradation profiles of all the three dyes, catalyzed Cu-doped TiO₂; the results are corrected for adsorption. The corresponding results with undoped TiO₂ are presented for comparison. Fig. 11 reveals the inhibition of photocatalytic degradation of the dyes on doping TiO₂ with copper by the described method. The results of *E. coli* inactivation under visible light and dye degradation under UV light clearly show the selectivity of Cu-doped TiO₂ for photocatalytic disinfection of bacteria.

3.4. Mechanism of photocatalytic disinfection

Band gap illumination of moist semiconductor generates reactive oxygen species such as HO·, O₂⁻, HO₂[·], and H₂O₂ [1,2], which may react with *E. coli*. However, the recently reported linear correlation between the steady-state concentration of HO· and the *E. coli* inactivation rate points to HO· as the most predominant species that takes up *E. coli* [32]. The photocatalytic bacteria disinfection is different from that of organic pollutant degradation. Some important differences are size, composition, viabilities, and inactivation process [33]. The size of organic molecules is less than 1 nm but *E. coli* is of micrometer size. It is of cylindrical shape of about 1.8 μm length and 0.5 μm diameter [18]. The observed Langmuir adsorption of the organics on the photocatalyst particles is not applicable to the microorganism owing to its size. *E. coli* are about 75 times larger than Cu-TiO₂ crystals. Hence it is the semiconduc-

tor that wraps the microorganism and not the other way. Also, the composition of a microorganism is very complex. Gram-negative bacteria such as *E. coli* possess 75-Å-thick outer cell membrane above the 25-Å-thick peptidoglycan layer on which about 35 million molecules of lipopolysaccharide (LPS) are present [33]. 75-Å-thick periplasmic space separates peptidoglycan and the inner 75-Å-thick cytoplasmic membrane. Periplasm inside the cell is covered by the cell fluid cytoplasm. Another important difference between the degradation of organic molecules and microorganisms is the viability of microorganisms [33]. Bacteria possess protection and recovery mechanisms to overcome the oxidative stress that involve endogenous proteins such as antioxidant enzymes or DNA repair enzymes. For example, enzymes such as Fe-SOD (SOD: superoxide dismutase), Mn-SOD, and catalase protect the cell from oxidative damage by catalyzing reactions that prevent the accumulation of O₂⁻ and H₂O₂ or exonuclease and DNA-glycoxylase enzymes repair DNA lesions resulting from oxidative damages. Yet another important point is the determination of photocatalytic efficiency. In the case of organic pollutants the

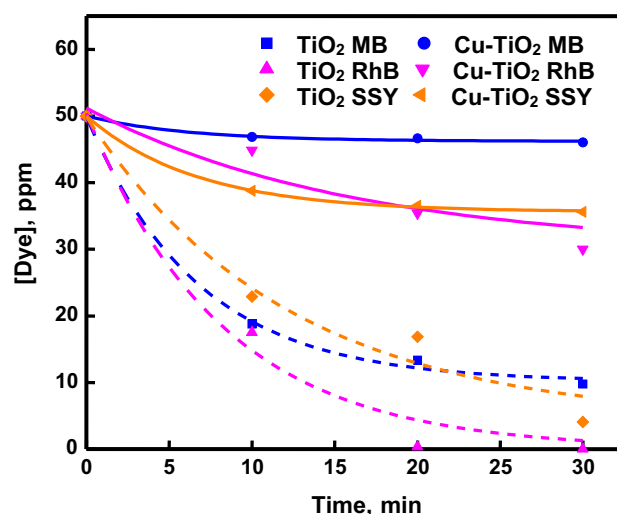


Fig. 11. Temporal profile of photocatalytic degradation of dyes under UV light. MB: methylene blue; RhB: rhodamine B; SSY: sunset yellow. Catalyst loading = 0.020 g, pH 5.5, airflow rate = 7.8 mL s⁻¹, [O₂]_{dissolved} = 9.3 mg L⁻¹, dye solution = 25 mL, λ = 365 nm, I = 25.4 μEinstein L⁻¹ s⁻¹.

photocatalytic efficiency corresponds to its disappearance and sometimes its mineralization. Only one modification in the structure of organic molecule leads to its disappearance. But the inactivation of a microorganism does not correspond to only one modification of an organic molecule present in the microorganism [33].

The TEM images presented in a few reports show aggregation of TiO₂ nanocrystals on the *E. coli* surface [34,35]. The reactive oxygen species induce damage to the cell membrane and the FE-SEM [18] and AFM [36] images displayed elsewhere reveal the damage to the smooth cell surface on illumination with TiO₂. The cell wall of *E. coli* acts as a barrier to the photokilling process [36]. Although the cell membrane damage is known to result in cell death, a mechanism to repair cell wall damage does exist and therefore bacteria cell wall damage alone will not cause bacterial inactivation. The changes in the concentration of the cell wall components during illumination show that while the outer membrane serves as a barrier, the peptidoglycan layer does not have a barrier function [36]. AFM images shown in a report reveal that the outer membrane decomposed first, and with further illumination, the cells completely decomposed [34]. The photokilling reaction is initiated by a partial decomposition of the outer membrane followed by dis-ordering of the cytoplasmic membrane, resulting in cell death. Further, ATR-FTIR study confirms the formation of peroxidation products due to TiO₂ photocatalysis [34]. The changes in the *E. coli* wall membrane are the precursor events leading to bacterial lysis. Moreover, the hole in the valence band may also grab an electron from CoA, resulting in dimeric CoA [33]. Dimerization of CoA inhibits the respiration and causes cell death.

A possible explanation for the observed very high photocatalytic bactericidal efficiency of Cu-doped TiO₂, compared to the N- or S-doped TiO₂ [23–26], is the effective binding of Cu–TiO₂ with *E. coli*. Copper(II) is more vulnerable to coordination with proteins than nitrogen or sulfur. Effective attachment of the nanocrystals on *E. coli* will lead to efficient flow of highly oxidative radicals from the illuminated semiconductor to *E. coli*. The EIS results reveal that the photocatalytic efficiency to inactivate *E. coli* is not determined by charge-transfer resistance, specific conductance, and capacitance of the composite oxide.

4. Conclusions

Cu–TiO₂ nanocrystals, prepared by a modified ammonia-evaporation-induced synthetic method and calcined at 450 °C, very efficiently catalyze the disinfection of *E. coli* under visible light. The doped oxide has been characterized by XRD, EDS, ESR, SEM, UV–visible DRS, PL, and EIS. Doping shifts the optical absorption edge to the visible region but increases the intragranular resistance and decreases the capacitance. The doped oxide is selective in photocatalysis; its photocatalytic activity under UV light to degrade sunset yellow, rhodamine B, and methylene blue dyes is less than that of bare TiO₂.

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